Data Mining Algorithms - Output Documentation

This document explains the specific outputs generated by each algorithm implemented in the AlgorithmsApp platform.

1. C4.5 Algorithm (Decision Tree)

- Output: Classification Decision Tree.
- Details: Generates a tree where each node represents an attribute split (including thresholds for numbers), branches represent attribute values, and leaves represent the final classification target.

2. ID3 Algorithm (Decision Tree)

- Output: Pure Categorical Decision Tree.
- Details: Similar to C4.5 but uses simple Entropy without thresholds. The output is a tree structure mapped directly from text categories.

3. PRISM Algorithm (Rule Induction)

- Output: Classification Rules (IF-THEN Format).
- Details: Extracts independent rules directly from data. Example: IF Outlook = Sunny AND Humidity = High THEN PlayTennis = No.

4. Naive Bayes (Probabilities)

- Output: Probabilistic Model (Prior & Conditional Probabilities).
- Details: Outputs a matrix of calculated probabilities showing the likelihood of each attribute occurring given a specific target class.

5. Apriori Algorithm (Association Rules)

- Output: Frequent Itemsets & Association Rules.
- Details: Outputs groups of items frequently bought together, followed by rules showing dependencies (Support % and Confidence %).

6. FP-Growth (Association Rules)

- Output: Frequent Itemsets & Association Rules.
- Details: Identical output format to Apriori, but extracted via an FP-Tree structure instead of candidate generation.

7. ECLAT (Association Rules)

- Output: Frequent Itemsets & Association Rules.
- Details: Identical output format to Apriori, but generated using vertical data intersections (TID Lists).